

|    |   |                                                                                                                                                                                                                                        |
|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | $(2R_E)^2 = \frac{4}{4}$ and new GPE becomes $-\frac{2R_E}{2} = -\frac{R_E}{2}$ , increased by $\frac{R_E}{2}$ .                                                                                                                       |
| 11 | B | <p>Gravitational force provides the centripetal force.</p> $F = \frac{GMm}{R^2} = m\left(\frac{R}{2}\right)\omega^2$ $2GM = R^3\left(\frac{2\pi}{T}\right)^2$ $\frac{2GM}{R^3} = \frac{4\pi^2}{T^2}$ $T = \sqrt{\frac{2R^3\pi^2}{GM}}$ |
| 12 | D | <p>Heat gained by A = Heat lost by B</p> <p><math>m \cdot c \cdot \Delta\theta = m \cdot c \cdot \Delta\theta</math></p>                                                                                                               |